Technological Innovation and Bursting Bubbles[1]

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Bubble and technological innovation

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- Examples:
  - 1720s French Mississippi bubble and British South Sea bubble: Atlantic trade, insurance
  - 1840s British railway mania: steam engine, railway network
  - 1890s British bicycle mania: pneumatic tire
  - 1920s U.S. stock price boom: electricity, consumer durables, automobile, etc.
  - 1990s U.S. dot-com bubble: Internet
  - Now: AI?

Rational asset price bubbles

- Bubble: asset price (Q) > fundamental value (V)
  - V = present value of dividends (D)
- Fundamental difficulty in generating asset price bubbles in real assets
  - Santos and Woodford (1997): bubble impossible if dividends nonnegligible relative to endowments
  - See Hirano and Toda (2024, JME) for illustration
- Theory of rational asset price bubbles attached to dividend-paying assets (including housing) largely underdeveloped
  - See Wilson (1981, JET), Hirano and Toda (2025a, JPE), Hirano and Toda (2025b, PNAS)

This paper

- Macro-finance model of innovation and stock bubble
  - Stock price (Q) $>$ fundamental value ($V :=$ PDV of dividends)
- Features:
  - Skilled agents choose to work in knowledge-intensive sector or establish new firms
  - Monopolistic competition: firm stocks pay dividends
  - Strength of knowledge spillover determines dividend growth rate
  - Agents expect spillover to eventually weaken (regime switching with absorbing state)

Main results

1. Agents rationally expect boom to eventually end, but bubble ($Q > V$) emerges as unique equilibrium outcome
   - Bubble necessity (Hirano and Toda, 2025a)
2. Long- and short-run effects of stock bubbles
   - Positive feedback between innovation and stock price
   - Despite inevitable collapse, bubble permanently increases output (because technology prevails) Bezos
   - Effect on wage inequality temporary
3. Implications for macro-financial modeling
   - Balanced growth is knife-edge (Uzawa, 1961; Schlicht, 2006)
   - Unbalanced growth and bubbles

Related literature

- Rational bubble: Samuelson (1958), Bewley (1980), Tirole (1985), Scheinkman and Weiss (1986), Kocherlakota (1992), Santos and Woodford (1997)
- Rational bubble attached to real assets: Hirano and Toda (2024, 2025a,b)
- Stochastic bubble: Blanchard (1979), Weil (1987)
- Technological innovation and asset boom: Olivier (2000), Pástor and Veronesi (2009)

Model: agents and preferences

- Time: $t = 0, 1, \ldots$
- Two period overlapping generations (OLG) model
- Aggregate uncertainty
- Epstein-Zin utility with unit EIS

$$U(c_t^y, c_{t+1}^o) = (1-\beta)\log c_t^y + \beta \log \mathrm{E}_t[(c_{t+1}^o)^{1-\gamma}]^{\frac{1}{1-\gamma}}$$

Model: endowments and dividends

- Young endowed with $e_t > 0$ units of good, old none
- Initial old endowed with unit supply of long-lived asset that pays dividend $D_t > 0$
- $\{(e_t, D_t)\}_{t=0}^{\infty}$ follows some stochastic process
- Budget constraints

$$\begin{aligned} &\text{Young:} && c_t^y + Q_t n_t = e_t, \\ &\text{Old:} && c_{t+1}^o = (Q_{t+1} + D_{t+1}) n_t, \end{aligned}$$

where Q_t: asset price, n_t: asset holdings

Equilibrium

Definition

Stochastic process $\left\{(Q_t, c_t^y, c_t^o, n_t)\right\}_{t=0}^{\infty}$ is *rational expectations equilibrium* if

1. (Utility maximization) initial old consume $c_0^o = Q_0 + D_0$; for each $t \geq 0$, (c_t^y, n_t, c_{t+1}^o) maximizes utility subject to budget constraints,
2. (Commodity market clearing) for each t, we have $c_t^y + c_t^o = e_t + D_t$,
3. (Asset market clearing) for each t, we have $n_t = 1$.

Unique equilibrium

- Due to unit EIS, optimal consumption of young is

$$c_t^y = (1-\beta)e_t$$

- Young budget constraint and $n_t = 1$ forces

$$Q_t = Q_t n_t = e_t - c_t^y = \beta e_t$$

Proposition

There exists unique rational expectations equilibrium. Asset price is $Q_t = \beta e_t$ and consumption is $(c_t^y, c_t^o) = ((1-\beta)e_t, \beta e_t + D_t)$.

Stochastic discount factor

- Let

$$m_{t\to t+1} = \frac{\partial U/\partial c^o_{t+1}}{\partial U/\partial c^y_t} = \frac{\beta}{1-\beta}\frac{c^y_t (c^o_{t+1})^{-\gamma}}{\mathrm{E}_t[(c^o_{t+1})^{1-\gamma}]}$$

  be stochastic discount factor (SDF) between time t and $t+1$

- Using equilibrium conditions,

$$\begin{aligned} m_{t\to t+1} &= \frac{\beta}{1-\beta}\frac{(1-\beta)e_t(\beta e_{t+1} + D_{t+1})^{-\gamma}}{\mathrm{E}_t[(\beta e_{t+1} + D_{t+1})^{1-\gamma}]} \\ &= \frac{Q_t(Q_{t+1} + D_{t+1})^{-\gamma}}{\mathrm{E}_t[(Q_{t+1} + D_{t+1})^{1-\gamma}]} \end{aligned}$$

- Depends only on asset price, dividend, and risk aversion

No-arbitrage condition

- Let $m_{t \to t+1}$ be SDF between t and $t+1$
- No-arbitrage condition is

$$Q_t = \mathrm{E}_t[m_{t \to t+1}(Q_{t+1} + D_{t+1})]$$

- Iteration yields

$$Q_0 = \mathrm{E}_0 \sum_{s=1}^{t} m_{0 \to s} D_s + \mathrm{E}_0[m_{0 \to t} Q_t],$$

where

$$m_{t \to t+s} := m_{t \to t+1} \times \cdots \times m_{t+s-1 \to t+s}$$

is SDF between t and $t+s$

Fundamental value and bubble

- Letting $t \to \infty$, get

$$Q_0 = \underbrace{\mathrm{E}_0 \sum_{s=1}^{\infty} m_{0\to s} D_s}_{=:V_0} + \underbrace{\lim_{t\to\infty} \mathrm{E}_0[m_{0\to t} Q_t]}_{=:B_0},$$

where

  - V_0: fundamental value,
  - B_0: bubble

- By definition, no bubble if and only if

$$\lim_{t\to\infty} \mathrm{E}_0[m_{0\to t} Q_t] = 0$$

Emergence of stochastic bubbles

- We now put more structure to derive stochastic bubbles

Assumption

There are two states denoted by u, b. Letting $z_t \in \{u, b\}$ denote state at time t, transition probabilities given by

$$\Pr[z_{t+1} = u \mid z_t = u] = \pi \in (0, 1),$$
$$\Pr[z_{t+1} = b \mid z_t = b] = 1.$$

- State u persists with probability π
- State b absorbing

State b exhibits balanced growth

Assumption

For any τ, conditional on $z_\tau = b$, sequence $\{(e_t, D_t)\}_{t=\tau}^{\infty}$ is deterministic and $e_{t+1}/e_t = D_{t+1}/D_t$ for all $t \geq \tau$.

State b exhibits balanced growth

Assumption

For any τ, conditional on $z_\tau = b$, sequence $\{(e_t, D_t)\}_{t=\tau}^{\infty}$ is deterministic and $e_{t+1}/e_t = D_{t+1}/D_t$ for all $t \geq \tau$.

Proposition

Once state b is reached, no bubble: $Q_t = V_t$.

State b exhibits balanced growth

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For any τ, conditional on $z_\tau = b$, sequence $\{(e_t, D_t)\}_{t=\tau}^{\infty}$ is deterministic and $e_{t+1}/e_t = D_{t+1}/D_t$ for all $t \geq \tau$.

Proposition

Once state b is reached, no bubble: $Q_t = V_t$.

- Intuition: $Q_t = \beta e_t$ grows with endowment
- In state b, uncertainty resolved and gross risk-free rate

$$R_{t+1} = \frac{\beta e_{t+1} + D_{t+1}}{\beta e_t} = \frac{e_{t+1}}{e_t}\left(1 + \frac{1}{\beta}\underbrace{\frac{D_{t+1}}{e_{t+1}}}_{\text{constant}}\right)$$

exceeds endowment growth, so discounting rules out bubbles

Condition for bubbles in state u

Assumption

Conditional on time $t-1$ information, endowment e_t and dividend D_t depend only on state $z_t \in \{u, b\}$.

Theorem

For $z \in \{u, b\}$, let (e_t^z, D_t^z) be value of (e_t, D_t) conditional on $z_0 = \cdots = z_{t-1} = u$ and $z_t = z$ and let $c_t^z := \beta e_t^z + D_t^z$. If $z_0 = u$, then there is a bubble at $t = 0$ if and only if

$$\sum_{t=1}^{\infty} D_t^u / e_t^u < \infty,$$

$$\sum_{t=1}^{\infty} (c_t^b / c_t^u)^{1-\gamma} < \infty.$$

Intuition and implications

1. Noting $Q_t = \beta e_t$, $\sum D_t^u / e_t^u < \infty$ implies $Q_t^u / D_t^u \to \infty$. Hence bubble can be understood as temporary deviation from balanced growth and explosive dynamics in P/D ratio
2. Equilibrium is unique. Hence (under these conditions) asset price bubble is necessity, not possibility
3. Conditions for stochastic bubbles stronger than deterministic case (Montrucchio, 2004, Proposition 7); if $\gamma < 1$, need crash to be larger the longer the bubble lasts

Model with innovation and intangible capital

- Endowment economy highly stylized
- We extend toy model to production, innovation, intangible capital (Grossman and Helpman, 1991)
  - R&D
  - Monopolistic competition

Agents and preferences

- Basically same as toy model
  - Two-period OLG model
  - Epstein-Zin preferences
- Mass $L > 0$ unskilled agents work in consumption good sector
- Mass $H > 0$ skilled agents either
  - Work in knowledge-intensive intermediate good firms, or
  - Engage in R&D and establish new firms

Consumption good sector

- Representative firm produces output (consumption good)

$$
\begin{aligned}
Y_t &= F(A_{Xt}X_t, A_{Lt}L_t) \\
&= \left(\alpha(A_{Xt}X_t)^{1-\rho} + (1-\alpha)(A_{Lt}L_t)^{1-\rho}\right)^{\frac{1}{1-\rho}}
\end{aligned}
$$

  where
  - X_t: knowledge-intensive good, L_t: unskilled labor
  - A_{Xt}, A_{Lt}: factor-augmenting productivities
  - $1/\rho < 1$: elasticity of substitution
- Maximizes profit $Y_t - P_tX_t - w_{Lt}L_t$, where
  - P_t: price of knowledge-intensive good
  - w_{Lt} unskilled wage
- Zero profit

Knowledge-intensive good sector

- Representative firm produces knowledge-intensive good

$$X_t = n_t^{1-1/\theta} \left(\int_0^{n_t} [x_t(j)]^\theta \, \mathrm{d}j \right)^{1/\theta},$$

where

  - n_t: "knowledge" (accumulates over time)
  - $x_t(j)$: knowledge-intensive intermediate good produced by firm j
  - $\theta \in (0, 1)$: elasticity parameter

- Maximizes profit

$$P_t X_t - \int_0^{n_t} p_t(j) x_t(j) \, \mathrm{d}j$$

- Zero profit

Knowledge-intensive intermediate good sector

- Intermediate goods differentiated by $j \in [0, n_t]$
- Skilled labor produces intermediate good $1:1$
- Firm j maximizes profit

$$d_t(j) = (p_t(j) - w_{Ht})x_t(j)$$

  by setting $p_t(j)$ (monopolistic competition), taking wage w_{Ht} and demand $x_t(j)$ as given
- Profit $d_t(j)$ paid as dividend to firm j stock

R&D sector

- New intermediate good varieties created through R&D
- 1 unit of skilled labor $\rightarrow$ an_t new varieties (firms)
- Founder sells stocks (claim to monoply profits) at IPO
- Hence indifference condition

$$w_{Ht} = Q_t an_t,$$

where $Q_t = q_t(j)$ stock price

Equilibrium

- First-order conditions of consumption good

$$P_t = F_X(A_{Xt}X_t, A_{Lt}L)A_{Xt},$$
$$w_{Lt} = F_L(A_{Xt}X_t, A_{Lt}L)A_{Lt}$$

- FOC of intermediate good j

$$p_t(j) = P_t(X_t/n_t)^{1-\theta}x_t(j)^{\theta-1}$$

- Hence demand for intermediate good j

$$x_t(j) = (X_t/n_t)(p_t(j)/P_t)^{-\frac{1}{1-\theta}}$$

Equilibrium

- Monopolistic competition implies

$$p_t(j) = w_{Ht}/\theta$$

- Because $p_t(j)$ common across j, so is $x_t(j) = x_t$
- Dividend

$$d_t(j) = (p_t(j) - w_{Ht})x_t(j) = \frac{1-\theta}{\theta} w_{Ht} x_t$$

  also common across j
- Hence may focus on symmetric equilibrium $Q_t = q_t(j)$

Optimal consumption and saving

- Skilled agents indifferent between working and R&D:

$$an_t Q_t = w_{Ht}$$

- Budget constraints of type $i \in \{H, L\}$

$$\begin{aligned} &\text{Young:} && c_{it}^y + Q_t n_{it} = w_{it}, \\ &\text{Old:} && c_{i,t+1}^o = (Q_{t+1} + D_{t+1}) n_{it}. \end{aligned}$$

- Optimal consumption $c_{it}^y = (1-\beta) w_{it}$, so stock demand

$$\begin{aligned} n_{Ht} &= \beta w_{Ht}/Q_t = \beta a n_t, \\ n_{Lt} &= \beta w_{Lt}/Q_t = (w_{Lt}/w_{Ht}) \beta a n_t. \end{aligned}$$

Market clearing

- Let $\phi_t \in (0, 1]$ be fraction of skilled agents working
- Market clearing for skilled labor:

$$X_t = n_t x_t = \phi_t H$$

- Fraction $1 - \phi_t$ engage in R&D, so

$$n_{t+1} = (1 + a(1 - \phi_t)H)n_t$$

- Stock market clearing:

$$\underbrace{n_{t+1}}_{\text{supply}} = \underbrace{Hn_{Ht} + Ln_{Lt}}_{\text{demand}}$$

Equilibrium

Proposition

There exists unique equilibrium. Letting $g(x) = (F_X/F_L)(x,1)$, ϕ_t *solves*

$$\frac{1}{aH} = \phi_t - 1 + \beta + \beta\left[\theta\frac{A_{Xt}H}{A_{Lt}L}g\left(\frac{A_{Xt}H}{A_{Lt}L}\phi_t\right)\right]^{-1}.$$

Equilibrium prices are

Knowledge-intensive good price:	$P_t = p_t(j) = F_X A_{Xt},$
Skilled wage:	$w_{Ht} = \theta F_X A_{Xt},$
Unskilled wage:	$w_{Lt} = F_L A_{Lt},$
Stock price:	$Q_t = \frac{w_{Ht}}{an_t} = \frac{\theta}{an_t} F_X A_{Xt},$

where F_X, F_L *are evaluated at* $(A_{Xt}H\phi_t, A_{Lt}L)$.

Production function and productivities

- Specialize production function and productivities (A_{Xt}, A_{Lt})
- Production function is CES:

$$F(X, L) = \left(\alpha X^{1-\rho} + (1-\alpha) L^{1-\rho}\right)^{\frac{1}{1-\rho}},$$

  where $\alpha \in (0, 1)$ and $1/\rho$ is elasticity of substitution
- As before, two states $z \in \{u, b\}$

Knowledge spillover

Assumption

Let there be two states indexed by $z \in \{u, b\}$ and $1/\rho < 1$ (complement). There exist constants $A_X, A_L > 0$ and $\xi_u, \xi_b, \lambda_u, \lambda_b \geq 0$ such that

$$(A_{Xt}, A_{Lt}) = (A_X n_t^{\xi_{z_t}}, A_L n_t^{\lambda_{z_t}}),$$

where

$$\xi_u > \lambda_u > \lambda_b = \xi_b.$$

- Spillover as in Frankel (1962) and Romer (1986)
- State b (absorbing) is balanced growth ($\xi_b = \lambda_b$)
- Spillover stronger in state u and in knowledge-intensive good sector

Equilibrium

- Equilibrium conditions

$$u: \quad \frac{1}{aH} = \phi_t - 1 + \beta + \frac{\beta(1-\alpha)}{\theta\alpha}\left(\frac{A_X H}{A_L L}\right)^{\rho-1} n_t^{(\xi_{z_t}-\lambda_{z_t})(\rho-1)}\phi_t^{\rho},$$

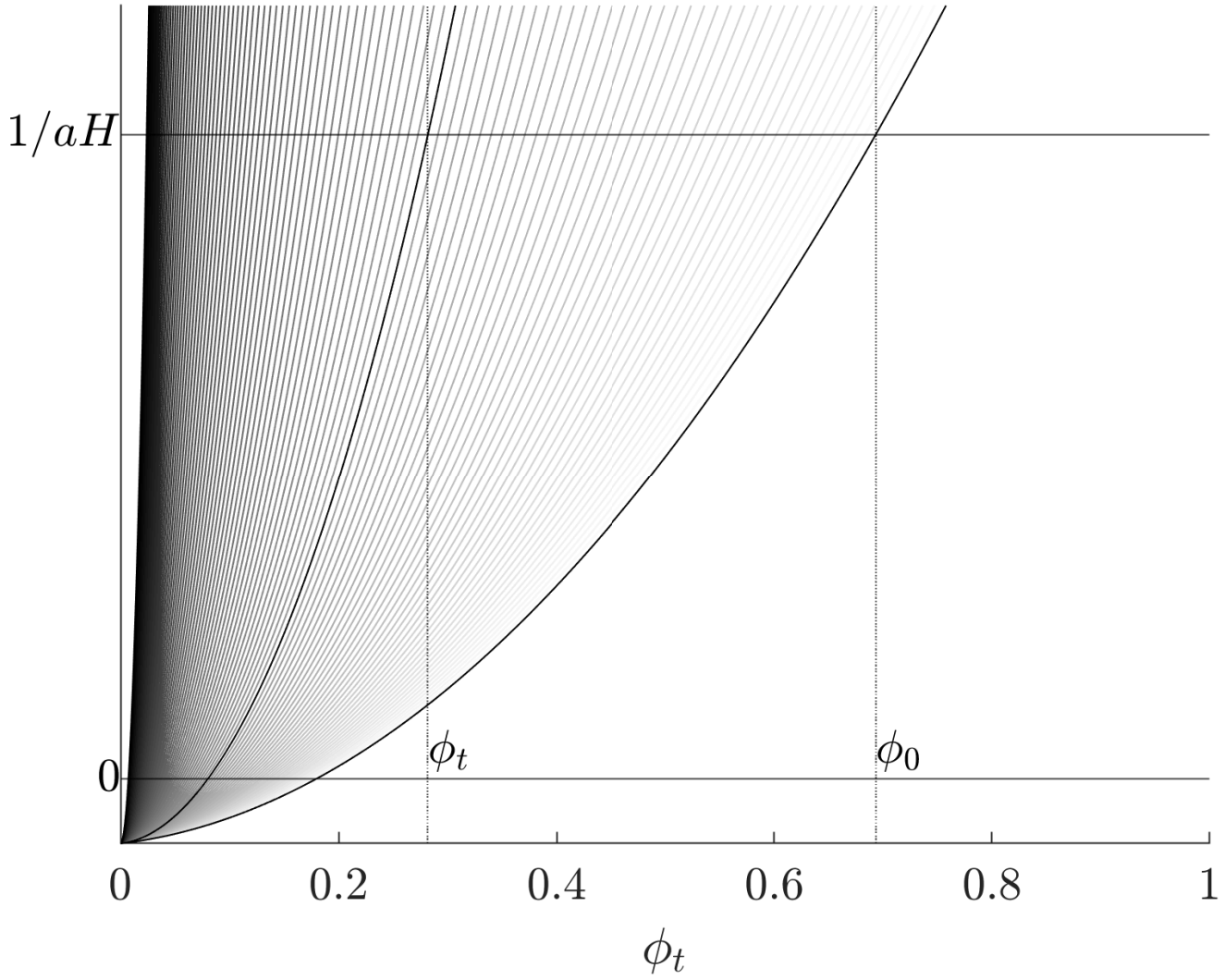
$$b: \quad \frac{1}{aH} = \phi_b - 1 + \beta + \frac{\beta(1-\alpha)}{\theta\alpha}\left(\frac{A_X H}{A_L L}\right)^{\rho-1} \phi_b^{\rho}.$$

Proposition

Under maintained assumptions, following statements are true.

1. *Conditional on staying in state u, $\{\phi_t\}$ monotonically converges to zero and knowledge n_t asymptotically grows at rate $G_u := 1 + aH$.*
2. *In state b, $\{\phi_t\}$ is constant at ϕ_b and knowledge n_t grows at rate $G_b := 1 + a(1-\phi_b)H < G_u$.*

Dynamics of ϕ_t



Inevitable emergence of stock price bubbles

- Equilibrium dynamics reduces to toy model

Theorem

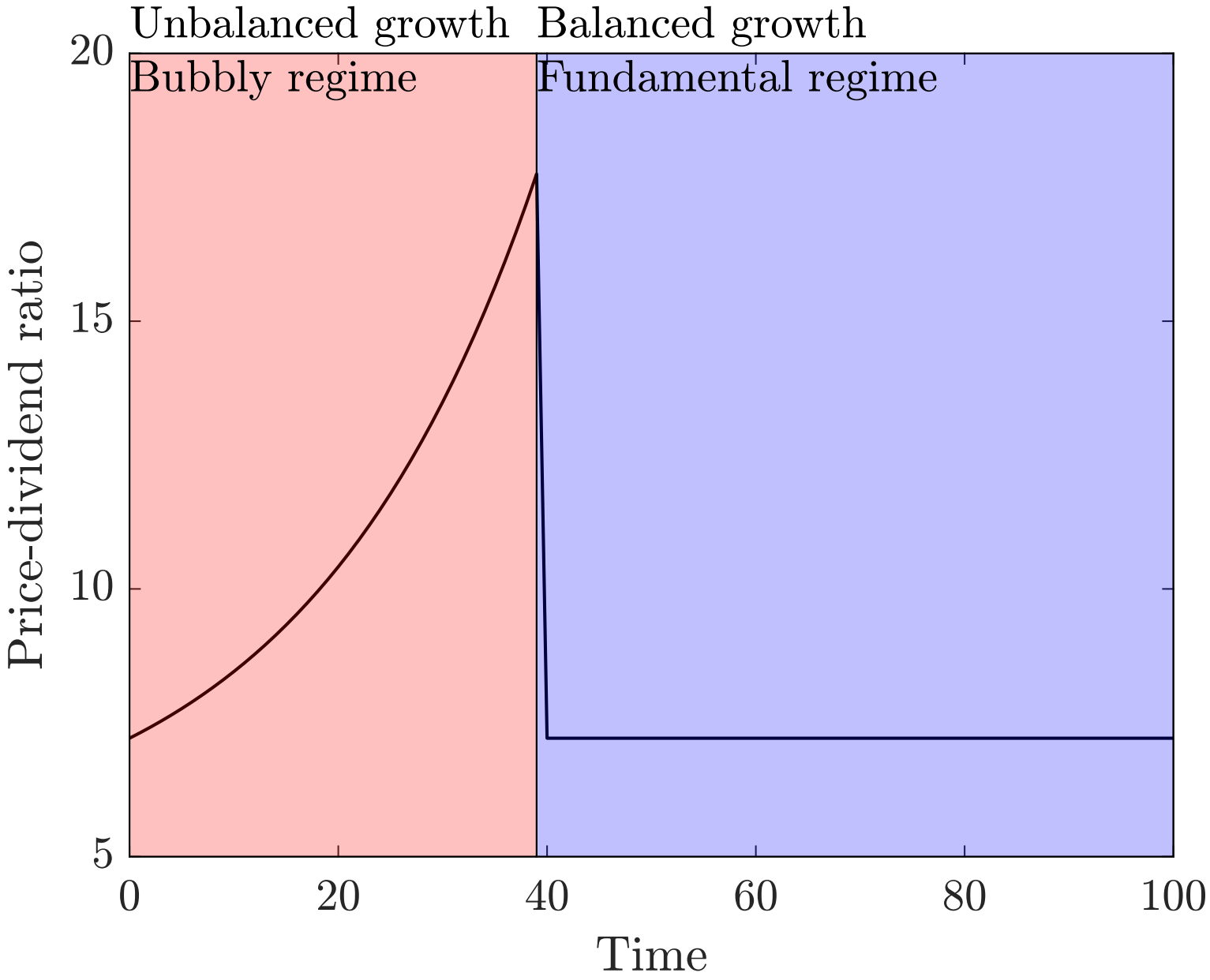
Suppose production function CES and relative risk aversion is $\gamma < 1$. Let Q_t be stock price in unique equilibrium and V_t fundamental value. Then

1. *In state $z_t = u$, stock price exhibits a bubble: $Q_t > V_t$ and price-dividend ratio Q_t/D_t grows exponentially.*
2. *In state $z_t = b$, stock price reflects fundamentals: $Q_t = V_t$ and price-dividend ratio Q_t/D_t is constant.*

Intuition and implications

1. Temporary unbalanced technological growth driven by regime switching and some conditions on elasticities necessarily generate stock price bubble
2. Dynamics of price-dividend ratio markedly different:
   - In state u, exponential growth
   - In state b, constant
3. Stock price bubble can be understood as temporary deviation from balanced growth; agents willing to buy overpriced stocks despite expecting collapse

Numerical example



Implication for skilled wage

- What is effect of stock price bubble?
- Stock price Q_t pushed above fundamental value V_t
- Indifference condition $w_{Ht} = an_t Q_t$, so $Q_t \uparrow \implies w_{Ht} \uparrow$

Observation

The stock market bubble tends to increase the skilled wage.

Implication for innovation

- Skilled wage $w_{Ht} = \theta F_X A_{Xt}$, where F evaluated at $(A_{Xt} H \phi_t, A_{Lt} L)$
- Noting F concave and (A_{Xt}, A_{Lt}) predetermined, $w_{Ht} \uparrow \implies \phi_t \downarrow$
- Hence fraction of skilled agents in R&D, $1 - \phi_t \uparrow$

Observation

The stock market bubble tends to promote innovation.

Implication for wage inequality

- Unskilled wage $w_{Lt} = F_L A_{Lt}$, where F evaluated at $(A_{Xt} H \phi_t, A_{Lt} L)$
- Hence $\phi_t \downarrow \implies w_{Lt} \downarrow$ (fixing n_t)

Observation

The stock market bubble tends to increase the wage gap between skilled and unskilled agents.

Implication for short-run output

- Assume state switches from u to b (bubble bursts) at t
- After burst, output is

$$Y_t = F(A_X H \phi_b, A_L L) n_t^{\lambda_b} \sim n_t^{\lambda_b}$$

- Before burst, output is

$$Y_{t-1} = F(A_{X,t-1} H \phi_t, A_{L,t-1} L) \sim n_{t-1}^{\lambda_u}$$

- Hence output growth has order of magnitude $n_t^{\lambda_b - \lambda_u}$

Observation

The longer the stock market bubble lasts (with higher n_t), the more severe the economic contraction when it bursts.

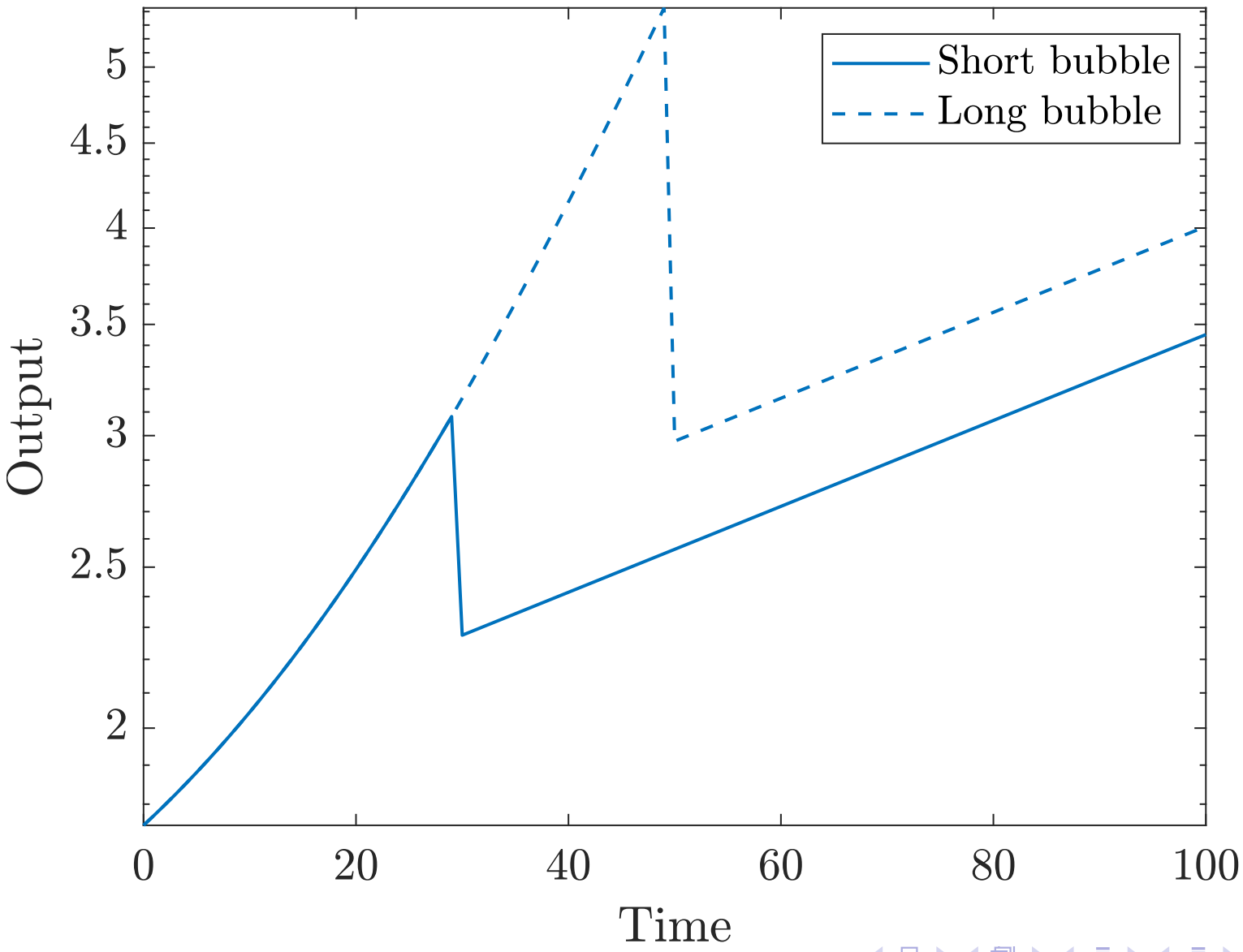
Implication for long-run output

- In long run, state is b and output $Y_t \sim n_t^{\lambda_b}$
- n_t larger (more innovation) the longer bubble lasts

Observation

The stock market bubble tends to increase the output in the long run.

Dynamics of output



Implication for long-run wages

- In long run, $\phi_t = \phi_b$ constant
- Hence relative wage

$$\frac{w_{Ht}}{w_{Lt}} = \theta \frac{F_X}{F_L} \frac{A_X}{A_L}$$

constant

Observation

The stock market bubble tends to increase wages in the long run but does not affect the wage gap between skilled and unskilled agents.

Balanced growth is knife-edge

- In macro, there is strong presupposition in balanced growth
- But balanced growth is knife-edge (Uzawa (1961) steady state growth theorem)

Proposition

Assume only Epstein-Zin utility and neoclassical production function F. Then price-dividend ratio Q_t/D_t is constant over time if and only if either relative productivity A_{Xt}/A_{Lt} is constant or production function F is Cobb-Douglas.
In particular, in our setting, parameters need to satisfy

$$\xi_u = \lambda_u \quad \text{or} \quad \rho = 1.$$

Conclusion

- Any balanced growth model is knife-edge theory
- Once we adopt unbalanced growth (here due to uneven technological spillover), asset price bubble becomes necessity
- Tight connection between technological innovation and stock bubble
- Innovation-driven stock bubble has many benefits (e.g., higher long-run output because more innovation) despite inevitable collapse

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TECH

Jeff Bezos says AI is in an industrial bubble but society will get 'gigantic' benefits from the tech

PUBLISHED FRI, OCT 3 2025•7:40 AM EDT | UPDATED FRI, OCT 3 2025•9:45 AM EDT

Arjun Kharpal
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"The [bubbles] that are industrial are not nearly as bad, it can even be good, because when the dust settles and you see who are the winners, societies benefits from those inventions." return

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